

Make the system agree with itself.

SYSTEM COHERENCE FRAME

Platform configuration

Compute + network

Sensors + actuation

Autonomy software

Validation evidence

Ownership + handoff

Executive readout

COMPANY MOMENT

Latitude is scaling a fully in-house autonomy platform toward a 2027 Ford vehicle launch, combining startup speed with automotive launch discipline.

ACTUAL MANDATE

Turn cross-domain ambiguity into a bounded problem, coherent repair, regression evidence, traceable configuration, and clear ownership closure.

CANDIDATE THESIS

Integration is not the phase after engineering. It is the mechanism that converts specialized velocity into one launchable system.

FIVE TENSIONS TO MANAGE

Startup speed vs launch discipline

Full-stack ownership vs domain authority

Unified architecture vs platform configuration

Architecture vs low-level debugging

Feature velocity vs regression stability

System Coherence Frame

FOUR MOVES THAT CLOSE THE ISSUE AND STRENGTHEN THE SYSTEM

01 FIND THE FRACTURE

Separate the visible symptom from the violated interface, invariant, timing condition, configuration, or ownership boundary.

02 ESTABLISH THE INVARIANT

Define the condition that must remain true across software, hardware, data, calibration, timing, and platform variants.

03 RESTORE THE COUPLING

Implement the smallest coherent repair while preserving domain authority and reviewing adjacent-system risk.

04 MAKE RECURRENCE HARDER

Leave behind a regression test, configuration check, observability, interface contract, or standard-work mechanism.

SCENARIO LOGIC

SCENARIO	FIRST USEFUL EXPERIMENT	CLOSURE EVIDENCE
New platform bring-up	Pin identities; reproduce on the smallest controlled platform slice.	Bounded matrix pass + traceable configuration.
Cross-domain regression	Bisect integrated changes while holding platform identity constant.	Failure before repair; stable result after repair.
Intermittent failure	Improve synchronized evidence and force the suspected boundary.	Known trigger with traceable before/after behavior.
Configuration drift	Diff software, firmware, calibration, network, and platform identity.	Equivalent declared configurations behave equivalently.

Evidence and technical value

VERIFIED EXPERIENCE MAPPED TO THE INTEGRATION MANDATE

Vape-Jet	Approximately 1,000 fielded software-enabled robots Connect machine, field, customer, support, quality, factory, engineering, and telemetry evidence.
ZeusVu	Regulated autonomous drone operations; 100% safety record Systems thinking across sensing, mapping, risk, authority, environment, and operating evidence.
Compunetics	Advanced electronics and high-reliability programs Root cause, process controls, quality systems, and engineering/manufacturing integration.
Amazon	24/7 launch mechanisms at high operating scale Visibility, escalation, standard work, prioritization, and mechanism building under pressure.
DudeWorth	Python, TensorFlow concepts, agentic workflows Software fluency, structured experimentation, system architecture, evaluation, and governance.

IMMEDIATE CONTRIBUTION

INTEGRATION TRIAGE Bound symptoms, hypotheses, and evidence gaps.	PYTHON TOOLING Build practical diagnostics and workflow automation.	SYSTEM SYNTHESIS Connect hardware, software, timing, configuration, and quality.
REGRESSION THINKING Convert closure into durable tests and controls.	TRACEABILITY Make platform and configuration lineage easier to trust.	CROSS-TEAM CLOSURE Clarify blockers, experiments, ownership, and release evidence.

Entry plan and interview agenda

FIRST 90 DAYS

0-30 | READ THE SYSTEM

Map platform/configuration, build, test, evidence, and blocker flows. Shadow debugging across domains and learn Latitude decision grammar.

31-60 | OWN BOUNDED CLOSURE

Take responsibility for a defined issue set. Improve reproduction, configuration traceability, and targeted regression coverage.

61-90 | STRENGTHEN THE MECHANISM

Lead one integration slice. Identify repeated boundary failures and encode closure into tests, configuration, or ownership mechanisms.

EXECUTIVE QUESTIONS

- 01 Where do integration blockers lose the most time today: reproduction, ownership, configuration identity, prioritization, or release evidence?
- 02 What distinguishes a strong integration closure from a defect that is merely no longer visible?
- 03 Which interfaces carry the greatest change burden as the unified platform approaches launch?
- 04 When should Autonomy Integration implement directly versus route work to the owning domain team?
- 05 What would demonstrate after six months that this engineer increased development velocity and confidence?
- 06 Which platform or configuration problem would be most useful as a technical case discussion?

CLOSING POSITION

Russell brings robotics operations, autonomous systems, high-reliability electronics, software fluency, root-cause discipline, and cross-functional execution. The value is turning cross-domain ambiguity into reproducible conditions, aligned ownership, and durable tests, configuration controls, and interface mechanisms.